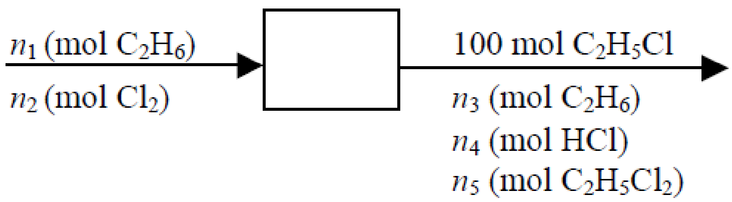
a. <u>Design for low conversion and feed ethane in excess.</u> Low conversion and excess ethane make the second reaction unlikely.

b. $C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl$, $C_2H_5Cl + Cl_2 \rightarrow C_2H_4Cl_2 + HCl$
<u>Basis</u>: 100 mol C_2H_5Cl produced



5 unknowns
<u>−3 atomic balances</u>
2 D.F.

c. <u>Selectivity:</u> $100 \text{ mol } C_2H_5Cl = 14n_5 \text{ (mol } C_2H_4Cl_2) \Rightarrow n_5 = 7.143 \text{ mol } C_2H_4Cl_2$

$$\left.\begin{array}{ll} \underline{\text{15\% conversion:}} & (1-0.15)n_1 = n_3 \\ \underline{\text{C balance:}} & 2n_1 = 2(100) + 2n_3 + 2(7.143) \end{array}\right\} \Rightarrow \begin{array}{l} n_1 = 714.3 \text{ mol } C_2H_6 \text{ in} \\ n_3 = 114.3 \text{ mol } C_2H_6 \text{ out} \end{array}$$

<u>H balance:</u> $6(714.3) = 5(100) + 6(114.3) + n_4 + 4(7.143) \Rightarrow n_4 = 607.1 \text{ mol HCl}$

<u>Cl balance:</u> $2n_2 = 100 + 607.1 + 2(7.143) \Rightarrow n_2 = 114.3 \text{ mol } Cl_2$

<u>Feed Ratio</u>: $114.3 \text{ mol } Cl_2 / 714.3 \text{ mol } C_2H_6 = \underline{\underline{0.16 \text{ mol } Cl_2 / \text{mol } C_2H_6}}$

<u>Maximum possible amount of C_2H_5Cl</u>:

$$n_{\max} = \frac{114.3 \text{ mol } Cl_2}{} \left| \frac{1 \text{ mol } C_2H_5Cl}{1 \text{ mol } Cl_2} \right. = 114.3 \text{ mol } C_2H_5Cl$$

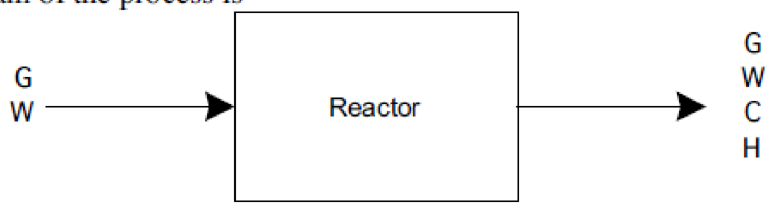
<u>Fractional yield of C_2H_5Cl</u>: $\dfrac{n_{C_2H_5Cl}}{n_{\max}} = \dfrac{100 \text{ mol}}{114.3 \text{ mol}} = 0.875$

d. Some of the $C_2H_4Cl_2$ is further chlorinated in an undesired side reaction:
$C_2H_5Cl_2 + Cl_2 \rightarrow C_2H_4Cl_3 + HCl$

P4.40
The balanced reaction is

$$C_3H_8O_3 + 3H_2O \leftrightarrow 3CO_2 + 7H_2$$

A flow diagram of the process is



where G = glycerol, W = water, C = carbon dioxide, and H = hydrogen.

There are 6 stream variables plus 1 system variable. We have 1 specified flow rate, 2 specified compositions, and 4 material balances, so DOF = (6 + 1) - (1 + 2 + 4) = 0.

The units are not consistent: the glycerol flow rate is given in g/h but the ratio of water:glycerol is given in molar units. We have a reaction with known stoichiometric coefficients, so we convert to units of gmol/h.

$$\dot{n}_{G,in} = \frac{150 \text{ g/h}}{92 \text{ g/gmol}} = 1.63 \text{ gmol/h}$$
$$\dot{n}_{W,in} = 5\dot{n}_{G,in} = 5 \times 1.63 = 8.15 \text{ gmol/h}$$

We write 4 material balance equations (all in units of gmol/h):

$$\dot{n}_{G,out} = \dot{n}_{G,in} - \dot{\xi} = 1.63 - \dot{\xi}$$
$$\dot{n}_{W,out} = \dot{n}_{W,in} - 3\dot{\xi} = 8.15 - 3\dot{\xi}$$
$$\dot{n}_{C,out} = 3\dot{\xi}$$
$$\dot{n}_{H,out} = 7\dot{\xi}$$

From the remaining stream composition specification, we write:

$$\frac{\dot{n}_{H,out}}{\dot{n}_{out}} = 0.54, \text{ where } \dot{n}_{out} = (1.63 + 8.15) - \dot{\xi} - 3\dot{\xi} + 3\dot{\xi} + 7\dot{\xi} = 9.78 + 6\dot{\xi}$$

Therefore, $\dfrac{\dot{n}_{H,out}}{\dot{n}_{out}} = 0.54 = \dfrac{7\dot{\xi}}{9.78 + 6\dot{\xi}}$, and $\dot{\xi} = 1.40$ gmol/h.

The fractional conversion of glycerol is: $f_{CG} = \dfrac{\dot{\xi}}{\dot{n}_{G,in}} = \dfrac{1.40}{1.63} = 0.86$

The equilibrium constant for this gas-phase reaction is

$$K_a = \frac{(y_C)^3 (y_H)^7}{(y_G)(y_W)^3} P^6 = \frac{(3\dot{\xi})^3 (7\dot{\xi})^7}{(1.63 - \dot{\xi})(8.15 - 3\dot{\xi})^3 (9.78 + 6\dot{\xi})^6} P^6$$

where we calculated the expression on the rhs from the material balances, recognizing that the mole fraction of a compound is simply the molar flow rate of that compound divided by the total molar flow rate. We can answer the question about whether the reactor outlet is at equilibrium in 2 ways:

1. The easy(er) way. Plug in the value $\dot{\xi} = 1.40$ gmol/h, obtained from our process flow calculations, and P = 1.2 atm, into the rhs of the equilibrium equation and compare the result to the known K_a. If less, then we are not at equilibrium. When I do this I calculate the rhs of the equilibrium expression to be 3.75, which is less than 55, so we are not at equilibrium.

2. The hard(er) way. Use the equilibrium expression above, with P = 1.2 atm, set K_a = 55, and solve for the extent of reaction at equilibrium. Compare to that calculated above. When I do this (which is done on a spreadsheet, and carefully, because there are multiple roots!), I find $\dot{\xi} = 1.574$ gmol/h. The equilibrium conversion is greater than the observed conversion, so the system is not at equilibrium.

From our solution method (2) we can calculate the fractional conversion of glycerol at equilibrium to be (1.574/1.63) or 0.97, and the mole fraction of hydrogen at equilibrium to be (7*1.574/(9.78+6*1.574)), or 0.57.

Reaching equilibrium at these reactor conditions will require increasing the residence time in the reactor. Increasing the reactor temperature might help, as usually the kinetics of reaction increase with temperature, but we don't know the effect of temperature on the equilibrium constant.

To increase the equilibrium conversion of glycerol, we can (1) decrease P or (2) increase the amount of water fed to the reactor. For example, if P = 1 atm, $\dot{\xi} = 1.60$ gmol/h. If the water:glycerol ratio increases to 10:1, $\dot{\xi} = 1.63$ gmol/h, and we get essentially 100% conversion. Of course we pay for this in that we need to process a lot more water.

P4.49

The reaction for the conventional process is

$$C_8H_{10} + 3O_2 \rightarrow C_8H_6O_4 + 2H_2O \qquad \text{(R1)}$$

Atom economy is $\dfrac{166}{106+3(32)} = 0.822$

The overall reaction for the proposed process is

$$C_8H_{10} + 6H_2O_2 \rightarrow C_8H_6O_4 + 8H_2O \qquad \text{(R2)}$$

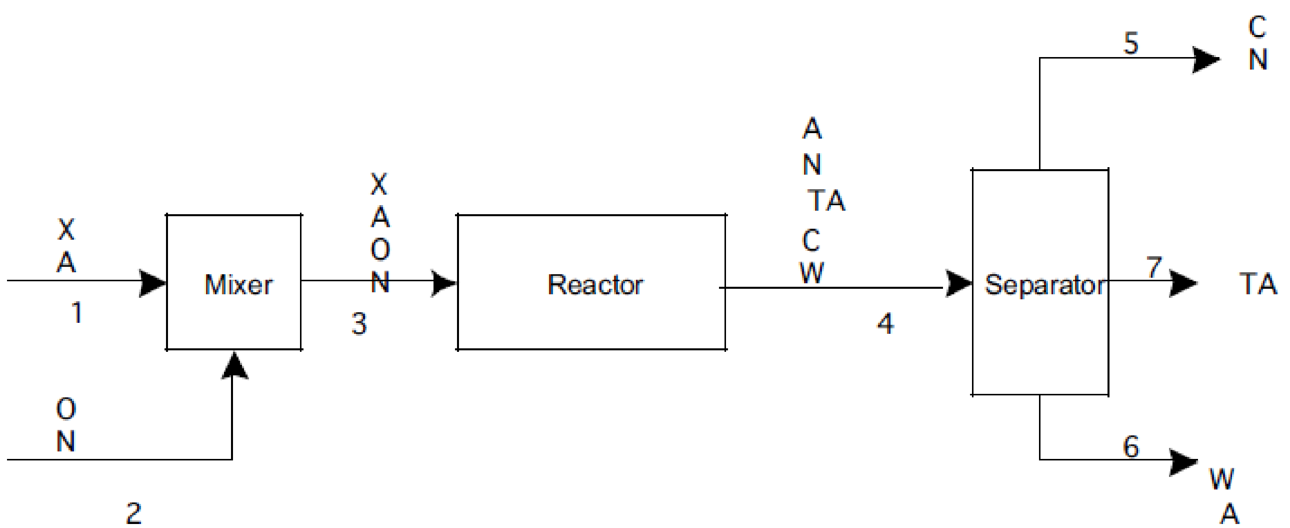
Atom economy is $\dfrac{166}{106+6(34)} = 0.535$ (not as good as conventional process).

Conventional process analysis:

In addition to (R1), oxidation of acetic acid occurs:

$$CH_3COOH + 2O_2 \rightarrow 2CO_2 + 2H_2O \qquad \text{(R3)}$$

The flow diagram (with X = xylene, O = oxygen, N = nitrogen, A = acetic acid, C = CO_2, W = H_2O, TA = terephthalic acid) is



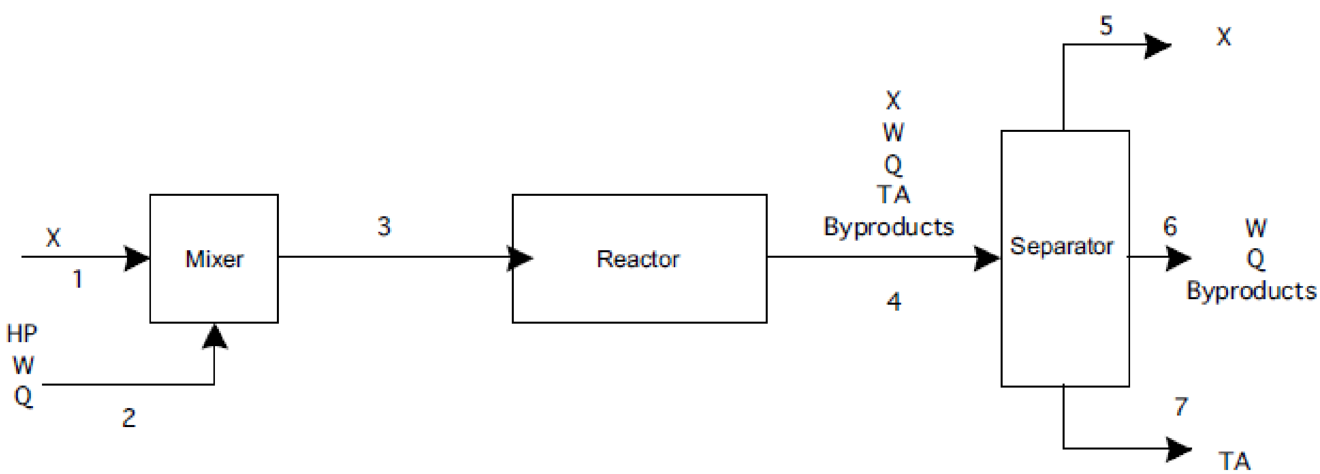
We'll assume (1) stream 1 is 10wt% acetic acid, (2) stream 2 is 21 mol% O_2, 79 mol% N_2 (3) complete conversion of xylene to TA, (4) 10% conversion of acetic acid to CO_2, (5) complete consumption of oxygen by reaction, and (5) perfect separators. The basis is 100 kg/day TA production, or 600 gmol/day TA. From the molar masses of the compounds, we convert stream 1 composition to 600 gmol/day xylene plus 15,000 gmol/day acetic acid. From these assumptions and the material balance equations at steady state, we calculate flows (all in gmol/day):

	1	2	3	4	5	6	7
X	600		600	0			
A	15000		13500	13500		13500	
O		4800	4800	0			
N		18048	18048		18048		
TA				600			600
C				3000	3000		
W				4200		4200	

(Notes: stoichiometric oxygen flow for R1 would be 3 x 600 or 1800 gmol/day. 10% conversion of acetic acid by R3 requires 2 x 0.1 x 15000 = 3000 gmol/day oxygen. The water produced comes from both (R1) and (R3).)

This process has excellent atom economy and yield. Its main drawback is that a significant amount of acetic acid is consumed. If the acetic acid is recycled (which would be necessary), the water would first need to be separated out. How difficult this separation is, remains to be seen.

Connie's process might look like this:



(Notes: Q is the inexpensive catalyst, HP is hydrogen peroxide. With a 70% yield and 90% selectivity, we can calculate that the fractional conversion of xylene is 0.778. For the same production rate of TA, we need to feed 600/0.7 or 857 gmol/day xylene. About 77.8%, or 667 gmol/day will react, of which 90% will react to the desired product PA. The desired reaction requires 6 x 600 or 3600 gmol/day HP; likely, the reaction will be run with some excess HP – I assumed 20% above this, and all consumed through either the desired or the side reactions. Water production will be 8 x 600 or 4800 gmol/day with the desired reaction, plus I estimated the extra HP all reacts to water, which adds 720 gmol/day water produced. We are not given information on how much supercritical water is added. The xylene would be recycled.)

The feasibility of Connie's process depends on (a) the nature of the byproducts – e.g., are they highly toxic, corrosive, or innocuous, (b) the difficulty of separating the unknown byproducts from X and TA, (c) the cost of the catalyst and the feasibility of recycling it or disposing of it, (d) the possibility for either improving the yield or recycling the xylene by tweaking reactor conditions, (e) the expense of hydrogen peroxide compared to air, and (f) the energy cost associated with making water supercritical. The key advantage of Connie's process is that enormous quantities of acetic acid are no longer required, however, there are many questions to be answered before one would get too enthusiastic about the alternative process compared to the tried-and-true.

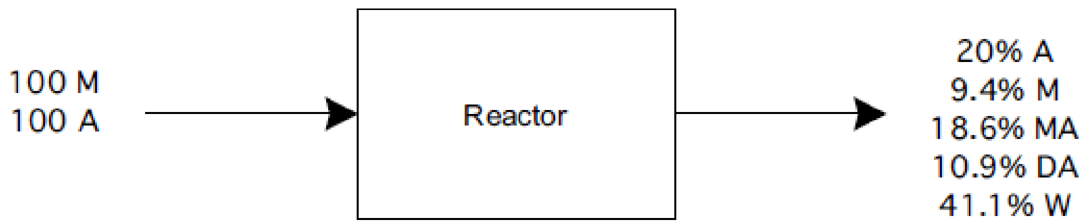
P4.50

The two balanced reactions are

$$NH_3 + CH_3OH \leftrightarrow CH_3NH_2 + H_2O \qquad (R1)$$

$$CH_3NH + CH_3OH \leftrightarrow (CH_3)_2NH + H_2O \qquad (R2)$$

The laboratory reactor flow diagram is shown.



where M = methanol, A = ammonia, MA = methylamine, DA = dimethylamine, and W = water. From the total mole balance equation at steady state:

$$\dot{n}_{out} = \sum_i \dot{n}_{i,in} + \sum_k \sum_i \nu_{ik}\dot{\xi}_k = \dot{n}_{M,in} + \dot{n}_{A,in} + \dot{\xi}_1 \sum_i \nu_{i1} + \dot{\xi}_2 \sum_i \nu_{i2}$$

$$= 100 + 100 + (-1 - 1 + 1 + 1)\dot{\xi}_1 + (-1 - 1 + 1 + 1)\dot{\xi}_2 = 200$$

Now we can proceed to check whether the data satisfy the element balance equations.

N:
$$100 = (0.20 + 0.186 + 0.109)200\,?$$
$$100 = 99\,? \quad \text{(close enough!)}$$

C:
$$100 = (0.094 + 0.186 + 2 \times 0.109)200\,?$$
$$100 = 99.6\,? \quad \text{(looks good!)}$$

H:
$$3 \times 100 + 4 \times 100 = (3 \times 0.2 + 4 \times 0.094 + 5 \times 0.186 + 7 \times 0.109 + 2 \times 0.411)200\,?$$
$$700 = 698\,? \quad \text{(yes!)}$$

O:
$$100 = (0.094 + 0.411)200\,?$$
$$100 = 101\,? \quad \text{(we're done!)}$$

The data all look excellent.

(a) fractional conversion of methanol $f_{CM} = \frac{\dot{n}_{M,in} - \dot{n}_{M,out}}{\dot{n}_{M,in}} = \frac{100 - 0.094 \times 200}{100} = 0.81$

(b) yield of methylamine based on methanol $y_{M \to MA} = \frac{0.186 \times 200}{100} = 0.37$

(c) selectivity of methylamine based on methanol $s_{M \to MA} = \frac{0.186 \times 200}{100 - (0.094 \times 200)} = 0.46$

(d) Ammonia participates in (R1) only so it is easiest to calculate the reaction rate r_1 using an ammonia material balance

$$\dot{n}_{A,out} = \dot{n}_{A,in} + \dot{r}_{A1}$$
$$0.20 \times 200 = 100 + \dot{r}_{A1}$$
$$\dot{r}_{A1} = -60 \text{ gmol/h}$$

Since the stoichiometric coefficient = 1 for all compounds, we have
$-\dot{r}_{A1} = -\dot{r}_{M1} = \dot{r}_{MA1} = \dot{r}_{W1} = \dot{\xi}_1 = 60$.
.

(e) Since dimethylamine participates in (R2) only:

$$\dot{n}_{DA,out} = 0.109 \times 200 = \dot{r}_{DA2} = 21.8 \text{ gmol/h}$$

Since the stoichiometric coefficient = 1 for all compounds, then
$\dot{r}_{DA2} = \dot{r}_{W2} = -\dot{r}_{MA2} = -\dot{r}_{M2} = \dot{\xi}_2 = 21.8$.

To check if the laboratory reactor is at equilibrium we calculate:

$$K_{a1} = \frac{y_{MA} y_W}{y_M y_A}$$
$$4 = \frac{(0.186)(0.411)}{(0.20)(0.094)}?$$
$$4 = 4.066 \text{ (close enough)}$$

$$K_{a2} = \frac{y_{DA} y_W}{y_M y_{MA}}$$
$$2.5 = \frac{(0.109)(0.411)}{(0.186)(0.094)}?$$
$$2.5 = 2.56 \text{ (close enough)}$$

Both calculations show that the reactor effluent is at equilibrium.

In the general case, the material balance equations tell us:

$$\dot{n}_{M,out} = \dot{n}_{M,in} - \dot{\xi}_1 - \dot{\xi}_2$$
$$\dot{n}_{A,out} = \dot{n}_{A,in} - \dot{\xi}_1$$
$$\dot{n}_{MA,out} = \dot{\xi}_1 - \dot{\xi}_2$$
$$\dot{n}_{DA,out} = \dot{\xi}_2$$
$$\dot{n}_{W,out} = \dot{\xi}_1 + \dot{\xi}_2$$

The equilibrium conversion can be calculated from

$$K_{a1} = 4 = \frac{y_{MA} y_W}{y_M y_A} = \frac{(\dot{\xi}_1 - \dot{\xi}_2)(\dot{\xi}_1 + \dot{\xi}_2)}{(\dot{n}_{M,in} - \dot{\xi}_1 - \dot{\xi}_2)(\dot{n}_{A,in} - \dot{\xi}_1)} \quad \text{and}$$

$$K_{a2} = 2.5 = \frac{y_{DA} y_W}{y_{MA} y_M} = \frac{\dot{\xi}_2(\dot{\xi}_1 + \dot{\xi}_2)}{(\dot{\xi}_1 - \dot{\xi}_2)(\dot{n}_{M,in} - \dot{\xi}_1 - \dot{\xi}_2)}$$

where the extents of reaction are those at equilibrium. Now we can use these equations to explore how changing the ammonia:methanol ratio affects yields and reactor outlet concentrations. We do this by fixing $\dot{n}_{A,in}$ and $\dot{n}_{M,in}$ at some desired flows, then solving the two equations simultaneously to find $\dot{\xi}_1$ and $\dot{\xi}_2$ (taking careful note that we find solutions that are physically reasonable), then calculate the desired quantities. Representative calculations are summarized in the table. All rates are in gmol/h.

$\dot{n}_{A,in}$	$\dot{n}_{M,in}$	$\dot{n}_{A,in}/\dot{n}_{M,in}$	$\dot{\xi}_1$	$\dot{\xi}_2$	% MA	MA/DA	$y_{A\to MA}$	$y_{M\to MA}$
100	50	2	36.8	8.1	19.0	3.5	0.29	0.57
100	100	1	59.4	21.8	18.8	1.7	0.376	0.376
100	200	0.5	83.2	47.4	11.9	0.75	0.36	0.18
100	400	0.25	95.9	74.0	4.4	0.29	0.22	0.055

Selectivity towards methylamine production, and yield of MA based on MeOH are best at excess ammonia. If methanol is expensive, separations are difficult, and there is no market for dimethylamine, we would likely choose to operate at excess ammonia with recycle. The yield of MA based on ammonia is best at a stoichiometric feed ratio. We might choose to operate at stoichiometric feed ratio if ammonia is expensive or we cannot recycle methanol. Operating at excess methanol does not make sense.